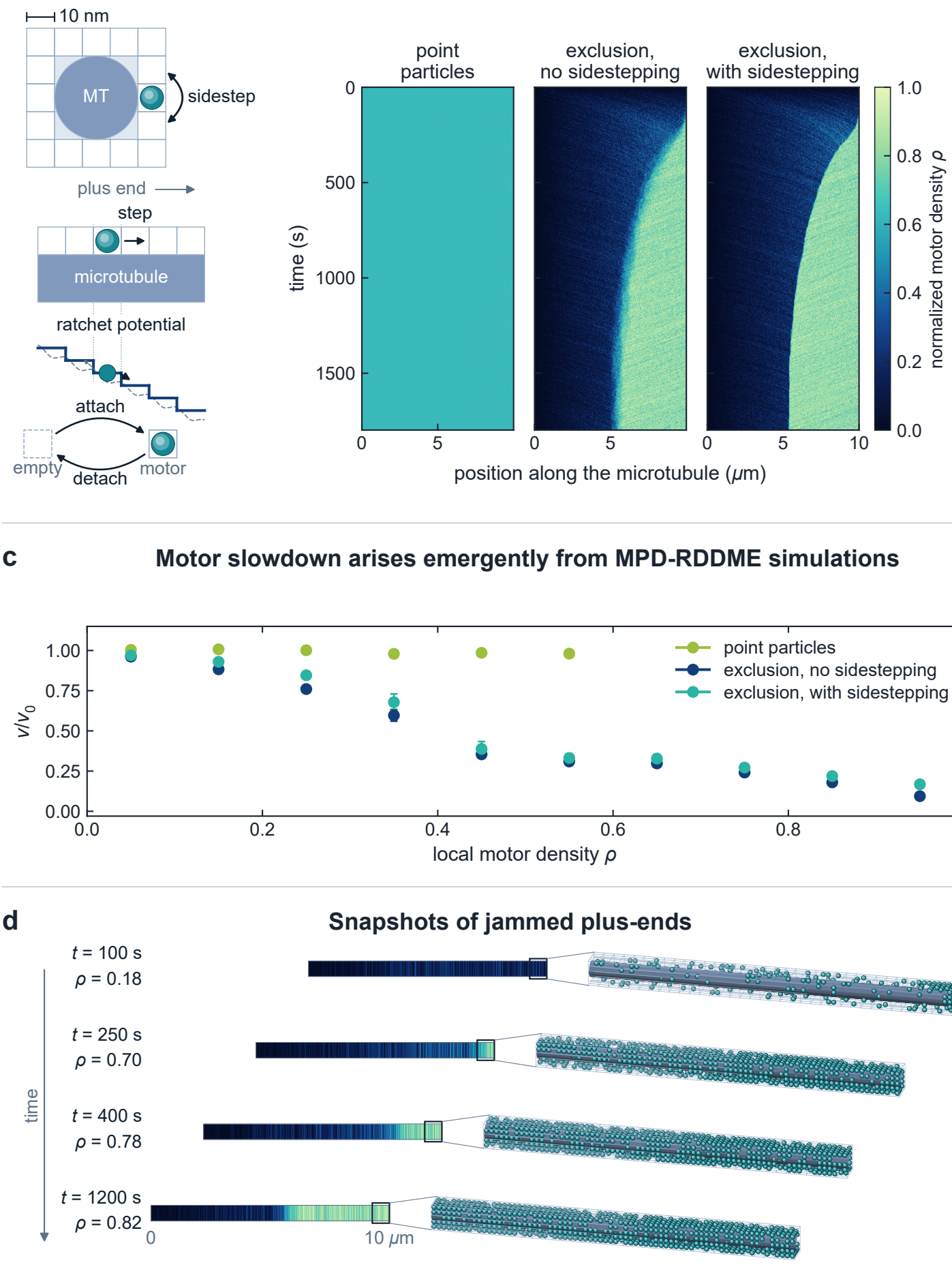

a Lattice model
10 nm
MT
sidestep
plus end
step
microtubule
ratchet potential
attach
empty
motor
detach
b Simulation of Kip3 traffic jams in microtubule plus-ends
point particles
exclusion, no sidestepping
exclusion, with sidestepping
time (s)
0
500
1000
1500
0
5
0
5
0
5
10
position along the microtubule (μm)
normalized motor density ρ
1.0
0.8
0.6
0.4
0.2
0.0
c Motor slowdown arises emergently from MPD-RDDME simulations
v/v0
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
local motor density ρ
point particles
exclusion, no sidestepping
exclusion, with sidestepping
d Snapshots of jammed plus-ends
t = 100 s
ρ = 0.18
t = 250 s
ρ = 0.70
t = 400 s
ρ = 0.78
t = 1200 s
ρ = 0.82
time
0
10 μm